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ARTICLE

Paleomagnetism of ~1.09 Ga Lake Shore Traps (Keweenaw Peninsula, Michigan): new results and implications

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Abstract: We report paleomagnetic data from a new section of the ~1.09 Ga Lake Shore Traps exposed on Silver Island (10 flows) and on the adjacent mainland (two flows) along the northwestern coastline of the Keweenaw Peninsula in Michigan. We also present new data from nine additional lava flows, sampled from the tip of the peninsula previously studied by Diehl and Haig in 1994. Samples from all these lava flows yield well-defined characteristic magnetization directions upon thermal demagnetization. After structural tilt correction, the directions from Silver Island (site-mean declination, $D = 276.9°$; site-mean inclination, $I = 44.4°$; 95% radius of confidence for site mean, $\alpha_{95} = 2.6°$; number of samples, $N = 10$) and mainland ($D = 298.7°$, $I = 36.0°$, $\alpha_{95} = 10.1°$, $N = 2$) flows are close to the directions from equivalent lava flows from the upper ($D = 277.8°$, $I = 41.0°$, $\alpha_{95} = 2.3°$, $N = 17$) and lower ($D = 300.0°$, $I = 34.9°$, $\alpha_{95} = 2.3°$, $N = 10$) sections of the middle Lake Shore Traps exposed at the eastern tip of the Peninsula, respectively. Testing the paleomagnetic directions for serial correlation shows that some of the sequential lava flows on Silver Island and from the middle Lake Shore Traps at the tip of the Peninsula record the same vector of the geomagnetic field. Combining these correlated directions yielded new mean directions for Silver Island ($D = 277.2°$, $I = 44.1°$, $\alpha_{95} = 3.1°$, $N = 8$), and the upper ($D = 277.0°$, $I = 40.4°$, $\alpha_{95} = 3.7°$, $N = 10$) and lower ($D = 298.6°$, $I = 33.3°$, $\alpha_{95} = 4.3°$, $N = 5$) middle Lake Shore Traps at the tip of the Peninsula. The statistical similarity of paleomagnetic directions obtained from these two locations with significantly different structural trends supports the conclusions of prior studies that the curvature of the Midcontinent Rift is primary. The new paleomagnetic pole for the Lake Shore Traps is located at 23.1°N, 186.4°E (95% confidence for the paleomagnetic pole, $A_{95} = 4.0°$; $N = 31$) and merits a nearly perfect six-point classification on the paleomagnetic reliability scale.

Résumé : Nous présentons des données paléomagnétiques d'une nouvelle section des trapps de Lake Shore, ~1,09 Ga, qui affleurent sur l'île Silver (dix coulées) et sur le continent adjacent (deux coulées) le long de la côte nord-ouest de la péninsule de Keweenaw, au Michigan. Nous présentons aussi de nouvelles données provenant de neuf autres coulées de lave, échantillonnées à la pointe de la péninsule, antérieurement étudiée par Diehl et Haig en 1994. Des échantillons de toutes ces coulées de lave ont donné des directions de magnétisation caractéristiques et bien définies lors de la démagnétisation thermique. Après avoir été corrigées pour l'inclinaison, les directions des coulées de l'île Silver (déclinaison moyenne au site, $D = 276.9°$; inclinaison moyenne au site, $I = 44.4°$; rayon de confiance de 95 % pour la moyenne du site, $\alpha_{95} = 2.6°$; nombre d'échantillons, $N = 10$) et du continent ($D = 298.7°$, $I = 36.0°$, $\alpha_{95} = 10.1°$, $N = 2$) se rapprochent des directions de coulées de lave équivalentes provenant respectivement de la section supérieure ($D = 277.8°$, $I = 41.0°$, $\alpha_{95} = 2.3°$, $N = 17$) et inférieure ($D = 300.0°$, $I = 34.9°$, $\alpha_{95} = 2.3°$, $N = 10$) des trapps centraux de Lake Shore qui affleurent à la pointe est de la péninsule. Des analyses des directions paléomagnétiques pour des corrélations de série montrent que quelques coulées de lave séquentielles sur l'île Silver et des trapps centraux de Lake Shore, affleurant à la pointe de la péninsule, enregistrent le même vecteur de champ géomagnétique. La combinaison de ces directions corrélées a donné de nouvelles directions moyennes pour l'île Silver ($D = 277.2°$, $I = 44.1°$, $\alpha_{95} = 3.1°$, $N = 8$) et pour le centre supérieur ($D = 277.0°$, $I = 40.4°$, $\alpha_{95} = 3.7°$, $N = 10$) et inférieur ($D = 298.6°$, $I = 33.3°$, $\alpha_{95} = 4.3°$, $N = 5$) des trapps de Lake Shore, à la pointe de la péninsule. La similitude statistique des directions paléomagnétiques obtenues de ces deux endroits, pourtant à tendances structurales très différentes, supporte les conclusions d'études antérieures stipulant que la courbature de la distension médio-continentale est primaire. Le nouveau pôle paléomagnétique pour les trapps de Lake Shore est situé à 23,1°N, 186,4°E (rayon de confiance de 95 % pour le pôle paléomagnétique, $A_{95} = 4.0°$, $N = 31$) et il se mérite une classification presque parfaite de six points sur l'échelle de fiabilité paléomagnétique. [Traduit par la Rédaction]

Introduction

The Lake Shore Traps (LST) is a sequence of thin lava flows interbedded within the Copper Harbor Formation (CHF) which crop out on the Keweenaw Peninsula in Michigan (Lane 1911). The LST represent the last significant stage of the ~1.1 Ga Midcontinent Rift (MCR) magmatism (Van Schmus et al. 1982). These rocks have been a subject of several paleomagnetic studies (e.g., DuBois 1962; Vincenz and Yaskawa 1968; Halls and Palmer 1981; Diehl and Haig 1994). In the most recent study, Diehl and Haig (1994) investigated 30 lava flows (sites) from three different stratigraphic sections within the CHF. The authors interpreted the high-coercivity magnetization component in these lava flows as primary. The site-mean directions corresponding to each package formed three distinct directional clusters (except for three intermediate directions). Based on low between-site angular dispersion within each cluster, Diehl and Haig (1994) concluded that each lava flow package had been emplaced over a short time period with respect to geomagnetic secular variation.

The main goal of this study was to extend the previous study by Diehl and Haig (1994) and to obtain a new high-quality paleomagnetic pole for LST by investigating a new section of 12 lava flows

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exposed on Silver Island and the adjacent mainland located between Eagle Harbor and Copper Harbor on the northwest coastline of the Keweenaw Peninsula (Fig. 1). We also sampled nine additional flows from an LST section at the tip of the Peninsula that were not sampled in the previous study of this section by Diehl and Haig (1994). Our new data combined with those from Diehl and Haig (1994) represent almost the entire Lake Shore Trap sequence. Here we report the results of our new paleomagnetic analyses and discuss their implications for the geological history of the Midcontinent Rift.

In particular, our new data allow us to examine whether the oroclinal bending of the Midcontinent Rift in the Lake Superior region is a primary feature or is at least in part a result of a later, post-rift deformation into an arcuate shape by a vertical axis rotation (Carey 1955). As pointed out by Hnat et al. (2006), the latter possibility is allowed by the fault inversion analysis (Witthuhn-Rolf 1997) and calcite-twinning analysis in basalts and sedimentary rocks (Craddock et al. 1997) that both find multiple paleostress directions attributed to rift closure. Furthermore, paleomagnetic data from sedimentary rocks from the western part of MCR in Wisconsin and Minnesota vary much more in declination than in inclination compared with those in Michigan, perhaps indicating a relative rotation between the two areas (Weil et al. 1998). Finally, we note that all modern analogs of MCR such as the presently active East African Rift or Baikal Rift do not manifest substantially curved segments. Therefore, investigation of the origin of MCR curvature may provide useful insight into the rift evolution and may potentially contribute to better understanding the kinematic evolution of deformational belts in general.

Paleomagnetism has proved an effective tool for investigating bending in orogenic belts (e.g., Eldredge et al. 1985; Sussman and Weil 2004). A regional fold test performed by Halls and Pesonen (1982) for MCR volcanic units showed a decrease in paleopole dispersion after applying structural tilt correction. This result suggests a pre-tilting magnetization for these volcanics and by inference no or minimal vertical axis rotation of sites with respect to each other. However, the problem of oroclinal bending was not discussed by Halls and Pesonen (1982). The problem of the origin of the MCR curvature was specifically addressed by Hnat et al. (2006) who tested the correlation between the in situ paleomagnetic declination and the lava flow strike (Eldredge et al. 1985) from 31 sites drilled in the ~1.1 Ga Portage Lake Volcanics. The sites were taken along the most curved part of the Keweenaw Peninsula to maximize the effect of structural trend variation. No correlation between the declination and strike was found, leading Hnat et al. (2006) to the conclusion that the MCR curvature was the primary feature probably reflecting the pre-existing areas of weakened lithosphere. However, because in Hnat et al.'s (2006) study, each site represented a separate lava flow, the declination–strike correlation could have been obscured by the changes in geomagnetic field direction due to secular variation. Here we re-examine this problem by comparing paleomagnetic directions from equivalent lava flows from two locations with a significant difference in the structural attitude.

The Midcontinent Rift rocks of the Keweenaw Peninsula

The ~1.1 Ga North American Midcontinent Rift system extends from northeast Kansas northward to Lake Superior and through Michigan and potentially northeast Ohio. Large volumes (>300 000 km^3) of primarily mafic, mantle-derived magma erupted in association with the evolution of the MCR (Green et al. 1987; Davis and Green 1997; Hutchinson et al. 1990). The Michigan segment of the MCR progressed from widespread localized magmatism during the early stage of rift to a main stage of magmatism within the down-dropped rift graben, which was followed by sedimentary infilling during post-magmatic sagging (Bornhorst and Lankton 2009). Compression resulting from the Grenville Orogeny at about 1.06 Ga caused the inversion of rift-bounding graben normal faults into high-angle reverse faults (Cannon 1994). During late rift compression, burial metamorphic hydrothermal fluids movement resulted in alteration of rift-filling rocks and formation of native copper deposits (White 1978; Bornhorst 1997).

Rocks of the main stage of rift magmatism and late rift compression are well exposed in the Keweenaw Peninsula of Michigan (Fig. 1). These rocks are characterized by a considerable variation in the dip direction along the strike of Peninsula, reflecting the curvature of the MCR in Lake Superior region. The dip directions vary from northwest at the western part of the Keweenaw Peninsula to almost completely north farther east and to northeast at the eastern tip of the Peninsula.

The oldest MCR unit exposed on the Keweenaw Peninsula is the Portage Lake Volcanics (PLV), a sequence of >200 lava flows with a total exposed thickness reaching almost 5300 m (White 1960). The Portage Lake Volcanics erupted within a 2–3 million year period around 1095 Ma (Davis and Paces 1990). Precise U–Pb zircon and baddeleyite ages of 1096.2 ± 1.8 and 1094.0 ± 1.5 Ma were obtained from the Copper City Flow, near the base of PLV, and from the Greenstone Flow, near the top, respectively (Davis and Paces 1990).

The Portage Lake Volcanics, in turn, interfingers with and is conformably overlain by the Copper Harbor Formation, with exposed thickness of up to 2000 m (Fig. 1). The Copper Harbor Formation includes a succession of lava flows, known as the Lake Shore Traps (Fig. 1). The Lake Shore Traps represents one of the latest magmatic events of the MCR (Van Schmus et al. 1982). A single lava flow from this sequence has been U–Pb dated at 1087 ± 1.6 Ma (Davis and Paces 1990). The flows of the LST are subaerial basalt and basaltic andesite, with thickness from 4.4 to 42.4 m, averaging 18.2 m (Paces and Bornhorst 1985). These lava flows display amygdaloidal flow bottoms, massive flow interiors, and vesicular flow tops typical of other Keweenawan subaerial flood basalts (Paces and Bornhorst 1985). Zeolite-grade alteration is common in permeable vesicular flow tops but is negligible or absent in flow interiors (Paces and Bornhorst 1985). The Lake Shore Traps are subdivided by conglomerate layers into three major groups: the lower, middle, and outer LST. The lower LST succession is not exposed and is identified only from magnetic surveys and drill-hole data (White et al. 1951).

The flows of the middle LST crop out at several locations along the Keweenaw Peninsula (Fig. 1). The most extensive outcrop of 31 flows, with the total thickness of about 600 m, is located at the eastern end (tip) of the Keweenaw Peninsula (Location 1, Fig. 1) where the flows dip to the northeast (022°), with a dip angle of ~24° (Diehl and Haig 1994). At Location 1, the middle LST is subdivided into two parts by a 27 m thick conglomerate layer, and following Diehl and Haig (1994), these are referred to as the lower and upper middle LST.

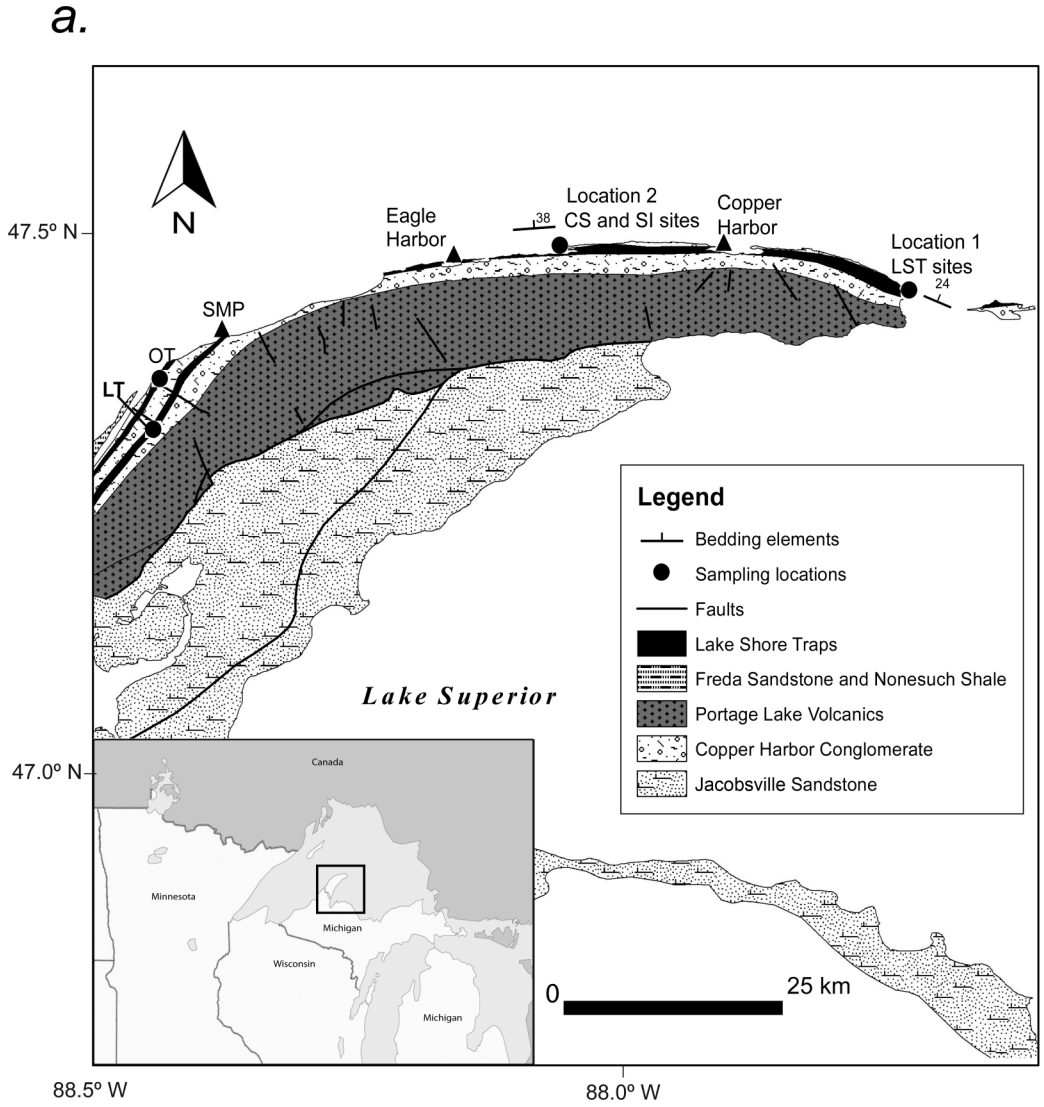
About ~30 km west from Location 1, 10 middle LST lava flows crop out on Silver Island and two additional flows are exposed on the mainland, directly across from the island (Location 2, Fig. 1). The lava flows dip at an angle of ~38° (Diehl et al. 2009), with a dip direction approximately north (356°). The number of exposed middle LST lava flows generally decreases westward. The westernmost exposure of the middle LST is represented by only two lava flows, each <20 m thick, separated by 30 m thick conglomerate layer (Halls and Palmer 1981). The outer LST succession consists of 8–14 lava flows located southwest of Seven Mile Point (Fig. 1).

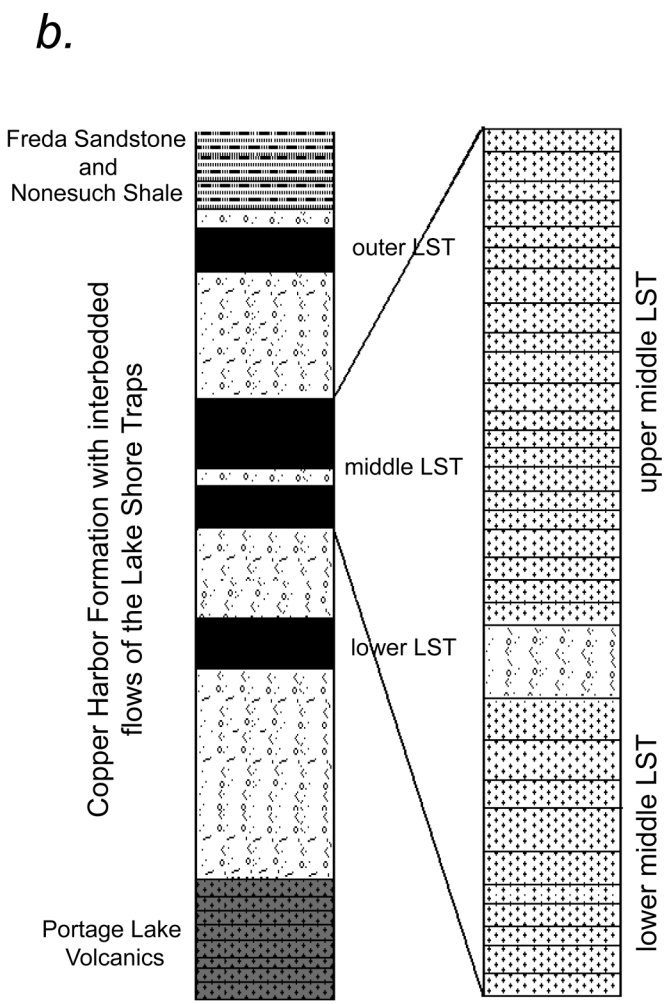
Previous paleomagnetic studies of the Lake Shore Traps

A comprehensive review of the previous paleomagnetic studies of LST is provided by Diehl and Haig (1994). The early paleomagnetic

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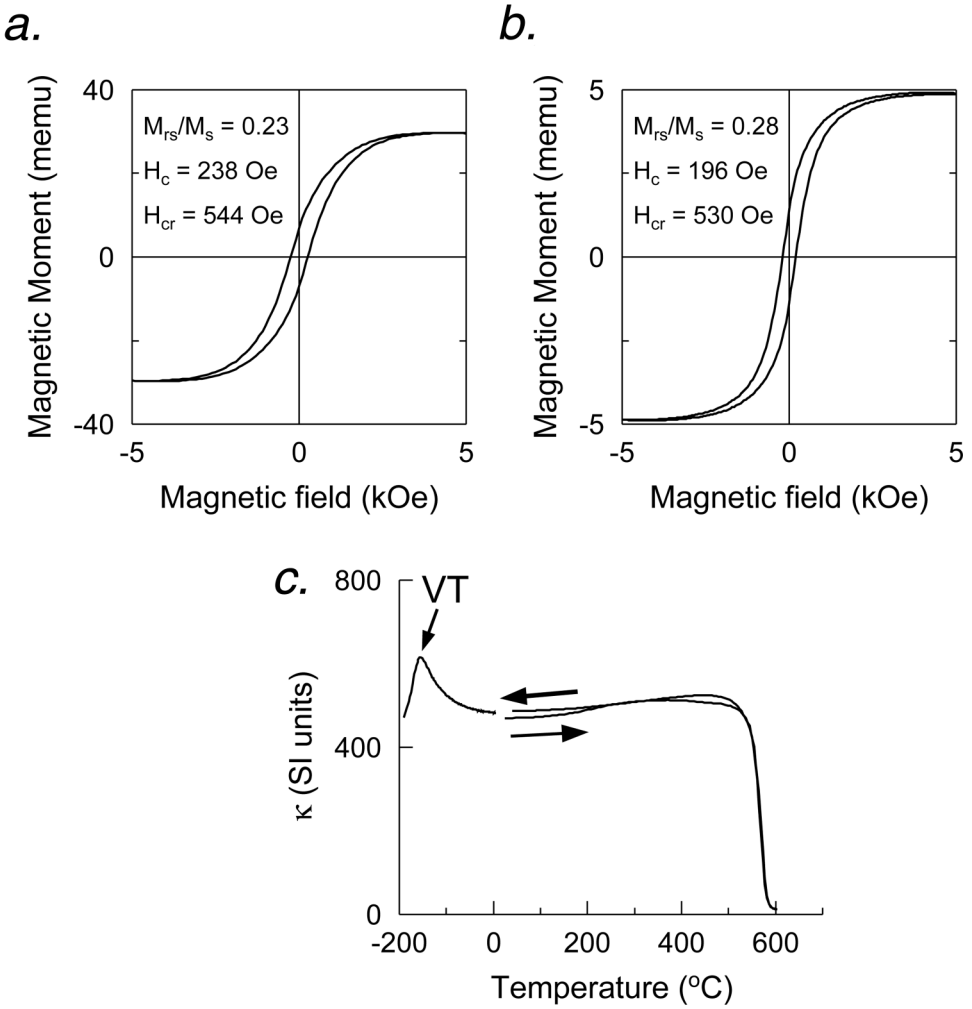
Fig. 1. (*a*) A simplified geologic map of the Keweenaw Peninsula showing our sampling Locations 1 and 2 of the middle Lake Shore Traps and the sampling locations for the outer LST (OT) and western middle LST (LT) of Diehl and Haig (1994). CS and SI are the coastal (mainland) sites and Silver Island sites, respectively (see text). SMP, Seven Mile Point. (*b*) A schematic stratigraphic column of rock units in the western portion of the Keweenaw Peninsula showing the distribution of the LST lava flow units within the Copper Harbor Formation (not to scale).





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Fig. 2. Rock magnetic properties of the middle LST sampled at Location 2. (*a, b*) Typical magnetic hysteresis loops for SI (*a*) and CS (*b*) sites. Abbreviations: M_{rs}, remanence magnetization; M_s, magnetization of saturation; H_c, coercivity field; H_{cr}, coercivity of remanence. (*c*) A typical dependence of low-field magnetic susceptibility (κ) versus temperature. The arrow shows the peak associated with the Verwey transition (VT), indicating the presence of nearly stoichiometric magnetite. 1 oersted (Oe) ≈ 79.6 A/m.



studies of the LST (DuBois 1962; Vincenz and Yaskawa 1968) were conducted on a small number of the middle and outer LST sites from the central portion of the Peninsula and did not utilize modern demagnetization and analytical techniques.

In a more comprehensive study, Halls and Palmer (1981) investigated eight sites of the middle LST distributed over ~50 km along the Peninsula's strike between the localities investigated in the earlier studies and Copper Harbor to the northeast. The northeastern sites yielded a characteristic remanent magnetization (ChRM) component with the mean direction (site-mean declination, D = 299.0°; site-mean inclination, I = 37.9°; 95% radius of confidence for site mean, α_{95} = 3.9°) statistically different from that (D = 282°, I = 12°, α_{95} = 12°) found by Vincenz and Yaskawa (1968). The primary origin of this ChRM was confirmed by a positive contact test (Palmer et al. 1981).

However, in the westerly sites, this direction was found to be intermediate between a soft recent-field overprint and two shallower high-coercivity components (D = 287.4°, I = 19.5° and D = 284.7°, I = 9.3°), which were interpreted as a chemical remanent magnetization (CRM) overprint caused by the mineralization event in the underlying PLV (Halls and Palmer 1981). Consequently, the authors also concluded that the shallow directions observed by Vincenz and Yaskawa (1968) reflect the same CRM overprinting.

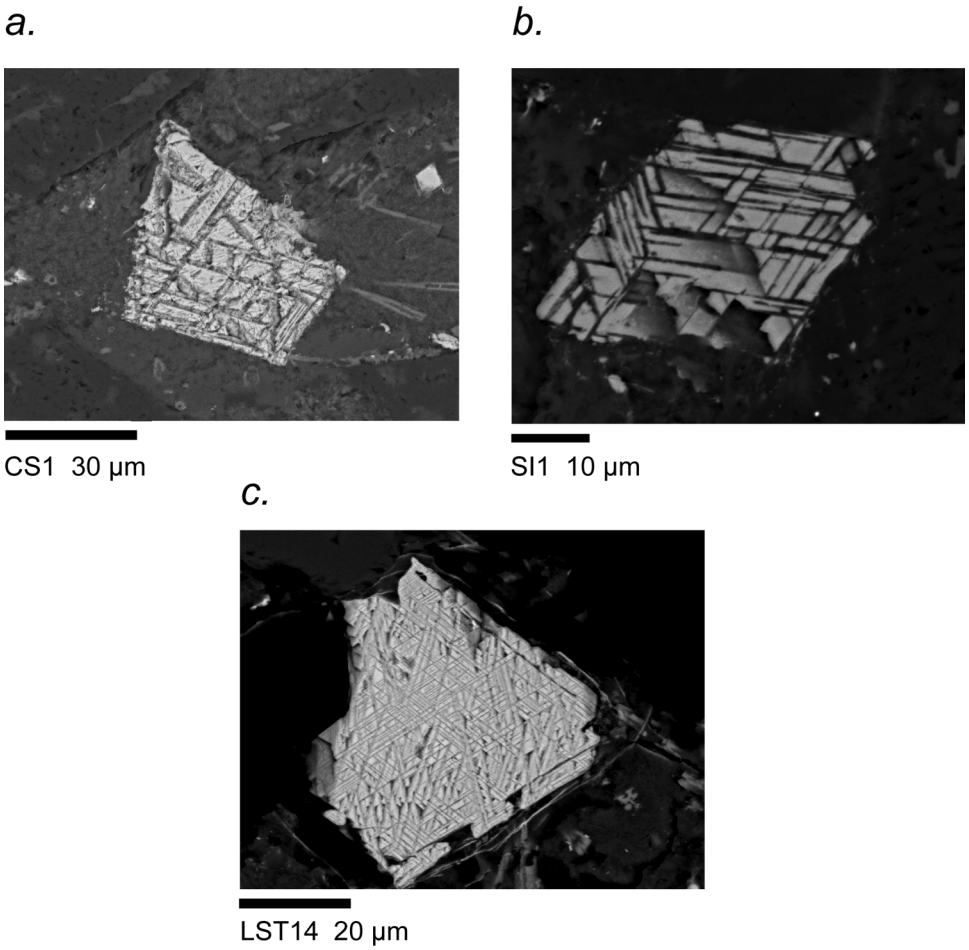
In the most recent study, Diehl and Haig (1994) investigated 30 LST sites (lava flows). Twenty flows of the middle LST were sampled at Location 1 (Fig. 1), with 10 flows sampled from the lower and upper middle LST sections each. Two additional middle LST sites were collected southwest of Seven Mile Point in the central part of the Peninsula (Site LT, Fig. 1). Eight other sites were collected from the outer LST along the Gratiot River in the central Peninsula (Site OT, Fig. 1).

All 30 sites yielded well-defined high-coercivity ChRM directions that were interpreted by Diehl and Haig (1994) as a primary magnetization. However, the site-mean directions from each stratigraphic lava flow package grouped into three distinct clusters with statistically different group-mean directions: D = 300.0°, I = 34.9°, α_{95} = 2.3°, number of studied sites N = 10 (the lower middle LST at Location 1); D = 279.3°, I = 38.7°, α_{95} = 4.9°, N = 10 (the upper middle LST at Location 1); and D = 277.8°, I = 5.2°, α_{95} = 3.0°, N = 8 (the outer LST). The two middle LST sites from Seven Mile Point area and one upper middle LST site resulted in intermediate directions (D = 295.4°, I = 12.9°, α_{95} = 2.4°), (D = 281.8°, I = 22.3°, α_{95} = 3.4°), and (D = 289.9°, I = 20.9°, α_{95} = 3.1°), respectively (Diehl and Haig, 1994 and fig. 7 therein).

Based on the low values of angular standard dispersion observed within each directional cluster, Diehl and Haig (1994) concluded that the corresponding lava flow packages were emplaced

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Fig. 3. Backscattered electron images of typical titanomagnetite grains of (*a*) CS, (*b*) SI, and (*c*) LST sample of the middle LST.



within short time periods representing spot readings of the geomagnetic field. Diehl and Haig further argued that the shallow and intermediate directions observed by Vincenz and Yaskawa (1968) and Halls and Palmer (1981) fall within the range of paleosecular variation defined by their more complete data; the discrepancy between the results of previous studies was hence interpreted as a result of incomplete sampling of paleosecular variation.

Results

Sampling

For this study, we collected 141 core samples from the middle LST at the two locations described earlier in the text (Fig. 1, Locations 1 and 2). At Location 1 (47.43°N, 87.71°W), we sampled nine flows (57 samples) of the upper middle LST that were not sampled in the previous study by Diehl and Haig (1994). In this paper, we refer to both previous and newly collected samples from this location as LST samples.

At Location 2 (47.46°N, 88.07°W), we collected 84 samples from 10 lava flows exposed at Silver Island (SI samples) and two lava flows exposed on the mainland (CS samples; CS, coastal sites). Five to eight core samples were collected from each lava flow at both locations. Cores were oriented using a magnetic compass checked by sun-compass readings.

Rock magnetism

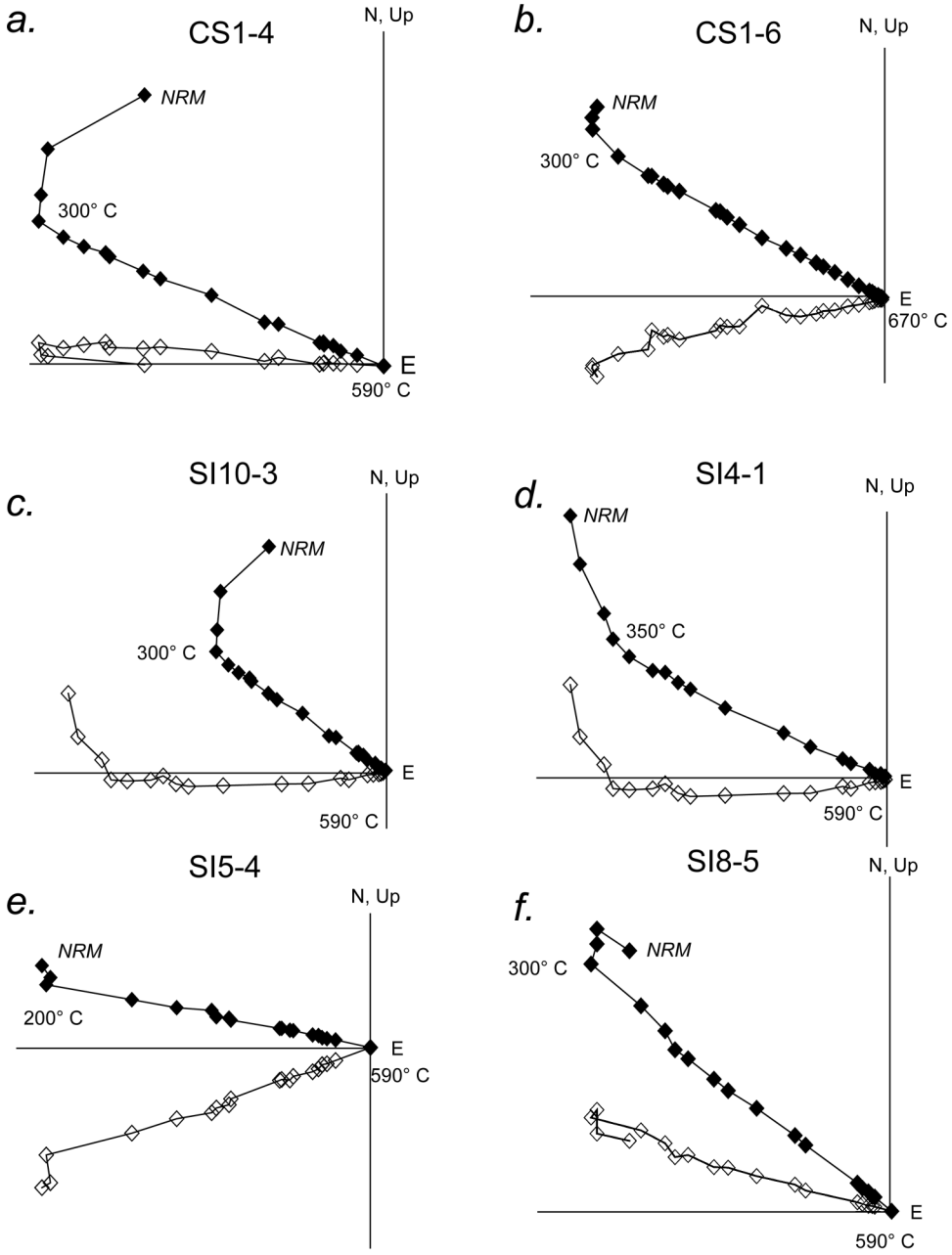
We conducted rock magnetic analyses on the middle LST flows sampled at Location 2 (samples SI and CS) to examine their magnetic mineralogy. Rock magnetic properties of the middle LST lava flows from Location 1 were previously reported by Diehl and Haig (1994).

Magnetic hysteresis parameters (coercivity, H_c; coercivity of remanence, H_{cr}; saturation remanence, M_{rs}; and saturation magnetization, M_s) were measured using a model 2900 Princeton measurement corporation alternating gradient field magnetometer. All measured magnetic hysteresis loops have a regular shape and indicate the presence of an intermediate coercivity magnetic phase (Figs. 2*a*, 2*b*). The M_{rs}/M_s and H_{cr}/H_c ratios suggest a pseudo-single domain (PSD) magnetic carrier in our samples (Day et al. 1977).

Temperature dependences of low-field magnetic susceptibility, $\kappa(T)$, were measured upon cycling from room temperature to 600 °C (in argon) using an AGICO MFK-1FA magnetic susceptibility meter equipped with a high-temperature furnace and a cryostat. The $\kappa(T)$ curves were also measured upon heating from –192 °C to room temperature before the high-temperature thermomagnetic runs. The thermomagnetic curves measured from both SI and CS samples are reversible, revealing the presence of a single magnetic phase, with a Curie temperature between 575 and 585 °C (Fig. 2*c*), consistent with magnetite and (or) Ti-low titanomagnetite as a magnetic remanence carrier. A characteristic peak observed at –153 °C, associated with the Verwey transition (Verwey 1939), indicates that the magnetite is nearly stoichiometric (Fig. 2*c*). Overall, the rock magnetic characteristics of the SI and CS flows are very similar to those reported by Diehl and Haig (1994) for the middle LST at Location 1, including reversible thermomagnetic curves and pseudo-single domain titanomagnetite with low Ti content as a predominant magnetic carrier.

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Fig. 4. Examples of vector end-point diagrams for the middle LST samples from Location 2 (open symbols are data point projections on horizontal plane; solid symbols are projections on vertical plane).



In addition to the rock magnetic investigation, we examined the opaque mineralogy of samples from both sampling locations (SI, CS, and LST) using a JOEL JSM-6400 scanning electron microscope (SEM) equipped with an energy dispersive spectrum (EDS) detector. Titanomagnetite grains were identified by means of backscattered electron imaging (using 20 KeV accelerating voltage and 9 mm working distance) and EDS analyses. Most identified grains were nearly equidimensional, subhedral, and ranging from ~20 to ~100 μm in size. The grains contained one or several subordinate sets of trellis type ilmenite exsolution lamellae in magnetite host (Haggerty 1991; Fig. 3).

Our SEM observations are consistent with the results of reflected light analyses by Diehl and Haig (1994) who observed ilmenite exsolution in all but one of their samples. Altogether, the reflected light, SEM, and rock magnetic analyses suggest that the middle LST lavas underwent high-temperature oxidation no higher that stage C3 (abundant ilmenite lamellae; equilibrium two-phase intergrowths) of Wilson and Watkins (1967) and Haggerty (1991).

Paleomagnetic results

Magnetic remanence measurements were conducted using a 2G Enterprises 760-R superconducting rock magnetometer. After measurement of their natural remanent magnetization (NRM), samples were cycled through the Verwey transition at ~120 K (Verwey 1939) by immersing them into liquid nitrogen to reduce a viscous magnetization component carried by larger magnetite grains (Schmidt 1993). Next, 14–18 thermal demagnetization steps were performed in an inert (nitrogen) atmosphere using an ASC TD-48SC thermal specimen demagnetizer. Progressive demagnetization was carried out until the magnetic intensity of the samples fell

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Table 1. Paleomagnetic results.

Site	N	D_g (°)	I_g (°)	D_s (°)	I_s (°)	α_{95} (°)	k	P_{lat} (°N)	P_{long} (°E)	A_{95} (°)	K
CS sites (Location 2)											
CS1	9	267.9	46.8	300.3	34.1	3.8	240	34.0	172.5	—	—
CS2	6	261.1	47.2	297.0	37.9	2.7	803	33.6	177.4	—	—
Mean	—	264.5	47.1	298.7	36.0	10.1	609	33.8	174.9	8.8	810
Lower middle LST sites (Location 1)											
LST1[a]	8	281.8	38.5	301.3	38.7	3.8	185	36.9	174.8	—	—
LST2[a]	7	283.5	38.0	302.5	37.6	4.8	163	37.1	173.2	—	—
LST3[a]	7	284.1	37.3	302.5	36.7	2.4	620	36.7	172.6	—	—
LST4[a]	8	281.9	36.4	300.1	36.8	2.1	681	35.2	174.6	—	—
LST5[a]	7	284.8	35.4	302.0	34.8	2.4	630	35.5	171.9	—	—
LST6[a]	7	283.3	35.0	300.4	35.1	3.4	228	34.5	173.4	—	—
LST7[a]	10	278.7	33.5	295.5	35.6	2.0	582	31.5	177.5	—	—
LST8[a]	8	282.6	34.8	299.7	35.2	3.9	207	34.1	174.0	—	—
LST9[a]	7	283.3	26.6	295.8	27.6	3.9	206	28.1	173.2	—	—
LST10[a]	7	286.0	31.3	300.7	30.7	2.1	863	32.7	170.8	—	—
Mean	—	283.0	34.7	300.0	34.9	2.3	426	34.3	173.6	1.9	639
SI sites (Location 2)											
SI1	9	239.2	43.5	280.4	48.1	3.0	282	27.7	195.8	—	—
SI2	6	237.2	37.8	272.4	45.8	5.0	153	21.2	199.5	—	—
SI3	6	243.9	40.5	279.9	43.5	4.0	281	24.9	193.1	—	—
SI4	8	236.6	40.9	275.5	48.1	2.3	1025	24.5	199.0	—	—
SI5	6	240.7	39.4	276.6	44.7	6.4	109	23.4	196.0	—	—
SI6	7	240.8	32.7	269.8	40.3	2.5	738	16.6	198.1	—	—
SI7	4	247.5	43.4	285.3	43.1	6.2	221	28.3	189.1	—	—
SI8	6	246.7	37.8	279.0	40.1	3.8	320	22.5	191.7	—	—
SI9	7	237.9	40.5	275.9	47.1	2.5	574	24.2	198.1	—	—
SI10	10	241.6	36.6	274.2	42.4	3.9	225	20.6	196.3	—	—
Mean	—	241.2	39.4	276.9	44.4	2.6	344	23.4	195.7	2.8	307
Upper middle LST sites (Location 1)											
LST11[a,b]	7	281.3	18.2	289.9	20.9	3.1	379	21.4	174.9	—	—
LST12	6	273.5	34.2	291.2	38.3	4.3	153	29.9	182.3	—	—
LST13[a]	8	266.3	35.5	285.3	42.4	4.5	155	27.9	189.1	—	—
LST14	6	260.0	36.4	279.7	45.7	4.1	160	25.9	195.0	—	—
LST15	6	262.9	34.2	281.1	42.6	3.2	142	25.2	192.1	—	—
LST16[a]	9	261.0	33.3	278.7	42.6	2.7	359	23.6	193.7	—	—
LST17[a]	8	260.4	34.5	278.9	43.9	3.3	292	24.4	194.4	—	—
LST18	7	263.5	32.8	280.8	41.1	6.6	85	24.2	191.4	—	—
LST19[a]	8	265.5	32.3	282.5	39.9	3.1	331	24.8	189.5	—	—
LST22	6	256.4	30.6	272.3	42.0	5.8	133	19.1	197.7	—	—
LST23[a]	8	260.0	32.1	276.9	41.9	2.0	786	22.1	194.5	—	—
LST24	6	257.0	24.2	269.4	36.0	2.5	481	14.2	196.5	—	—
LST25	6	260.6	34.6	279.1	43.9	6.0	104	24.6	194.2	—	—
LST26[a]	8	256.1	30.4	271.9	41.9	5.5	104	18.8	197.9	—	—
LST28	8	262.7	27.5	276.9	36.8	4.9	159	19.5	191.7	—	—
LST29[a]	9	260.8	29.7	276.2	39.5	2.1	593	20.4	193.6	—	—
LST30	6	255.9	24.4	268.4	36.6	5.1	144	13.8	197.5	—	—
LST31[a]	8	258.8	29.6	274.2	40.2	3.9	208	19.4	195.4	—	—
Mean	—	261.2	31.6	277.8	41.0	2.3	244	22.3	193.4	2.5	205

Note: N is the number of samples per site; D_g, I_g and D_s, I_s are the site-mean declination and inclination in geographic and stratigraphic coordinates, respectively; α_{95} is the 95% radius of confidence; k is the Fisher (1953) precision parameter; P_{lat} and P_{long} are the latitude and longitude of a virtual geomagnetic pole or a group-mean paleomagnetic pole. A_{95} and K are the 95% radius of confidence and the precision parameter for the group-mean paleomagnetic pole, respectively. Dip azimuth/angle used for structural correction for the structural tilt are 22°/24° for the middle LST flows at the tip of the Keweenaw Peninsula (Location 1) and 356°/38° for the flows of Silver Island (SI) and the mainland (CS) (Location 2).

[a]Paleomagnetic directions from Diehl and Haig (1994).

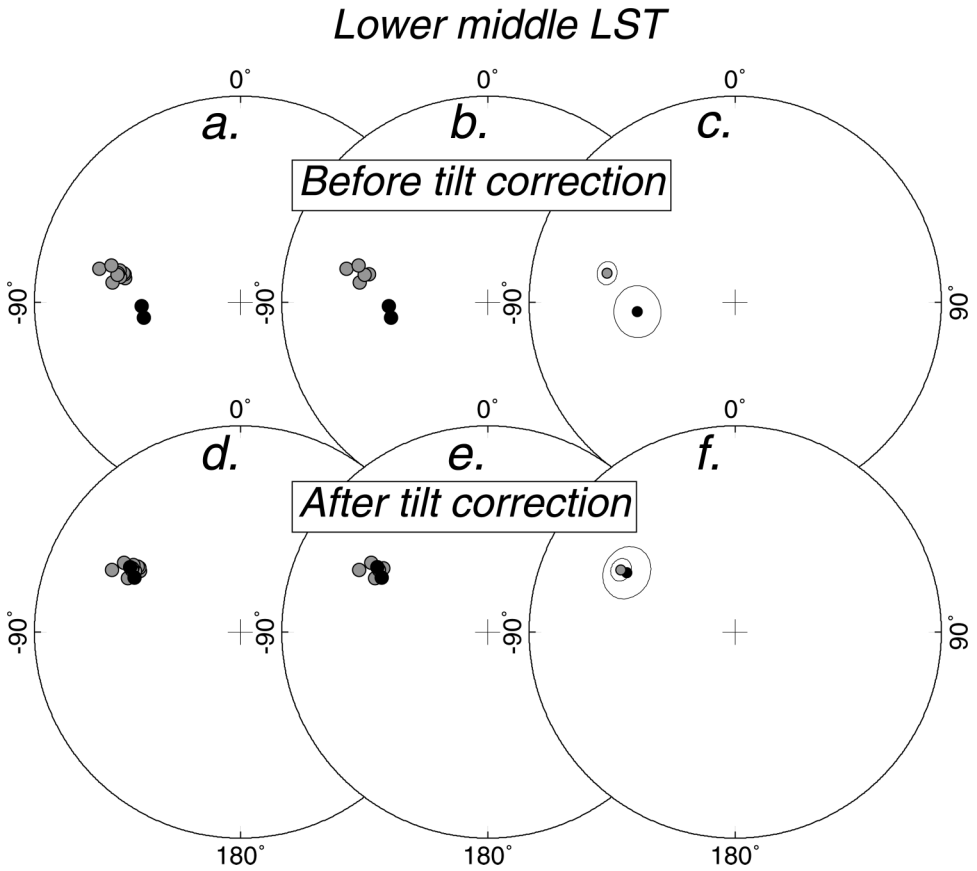
[b]Directions not used in calculations.

below noise level or until the measured directions became erratic and unstable (typically at 560–590 °C).

Nearly all CS, SI, and LST samples manifested a two-component NRM. A low-temperature, soft component was removed by heating not exceeding 300 °C for the CS and SI samples and 375° C for the upper middle LST samples. The characteristic remanent magnetization component (ChRM) was identified by straight linear demagnetization trajectory toward the origin on a vector endpoint diagram. The ChRM was typically demagnetized (to <5% of its initial value) by 560–590 °C (Fig. 4). In addition, samples from nine flows of the upper middle LST yielded a high-temperature (590–670 °C) component (e.g., Fig. 4*b*), most likely carried by hematite. The direction of this component is indistinguishable from that carried by magnetite, suggesting the deuteric nature of hematite formed during the original cooling of lavas. The ChRM directions were calculated using principal-component analysis. The best-fit line was used if the line was defined by at least six consecutive demagnetization steps that trended toward the ori-

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Fig. 5. Paleomagnetic site-mean directions determined from the lower middle LST sites at Location 1 (grey circles) and the CS sites at Location 2 (black circles) before (*a*, *d*) and after (*b*, *e*) combining serially correlated site-mean directions (see text). (*c*, *f*) The group-mean directions and their respective 95% confidence areas (α_{95}) based on the data in *b* and *e* (Table 2). The data are plotted (equal area projection) before (*a*–*c*) and after (*d*–*f*) correction for structural tilt.



gin and had a maximum angle of deviation (MAD) <10°. The mean MAD of the accepted ChRM was 1.4°, with the maximum MAD of 4.2°. The site- and group-mean directions were calculated using Fisher statistics (Fisher 1953). The mean for a site was accepted if the mean direction was obtained from three or more samples, the confidence circle (α_{95}) was smaller than 15°, and the precision parameter *k* was >50. Data from all our sites met these acceptance criteria.

The tilt-corrected site-mean directions from the two CS sites (Table 1) are similar to the direction (D = 292°, I = 38°) reported for the positive baked contact test by Palmer et al. (1981), hence supporting their primary nature. Furthermore, the CS mean direction (D = 298.7°, I = 36.0°, α_{95} = 10.1°, N = 2) is statistically indistinguishable from the group mean (D = 300.0°, I = 34.9°, α_{95} = 2.3°, N = 10) for the lower middle LST flows at Location 1 (Diehl and Haig 1994; Fig. 5; Table 1), with the caveat that the CS group mean is only based on two sites.

To further test the similarity of CS directions with the directions from the lower middle LST at Location 1, we applied the "isolated observation" test of McFadden (1982). In this test, the critical angle γ_0 for a given probability is compared with the angle between the isolated direction and the mean of N directions on the other group. If the calculated angle exceeds γ_0, the isolated observation should be treated as discordant. However, the angles calculated for the CS1 and CS2 site-mean directions (0.8° and 3.9°, respectively) do not exceed the critical angle γ_0 = 7.7° calculated for the lower middle LST site-mean directions (N = 10) at the 95% probability. Therefore, at 95% confidence, the CS directions belong to the same distribution as the lower middle LST, indicating that the CS flows belong to the same stratigraphic package.

On the other hand, the tilt-corrected group mean for SI sites (D = 276.9°, I = 44.4°, α_{95} = 2.6°, N = 10) is similar to that calculated for the upper middle LST flows exposed at Location 1 (D = 277.8°, I = 41.0°, α_{95} = 2.3°, N = 17). However, while the α_{95} circles for the group means overlap, neither circle of confidence contains the mean direction from the other group. Nevertheless, the maximum *k* statistic for the combined directions from both locations (McFadden and Jones 1981) was achieved at 100% unfolding. The general similarity of the directions indicates that the SI flows belong to the upper middle LST package.

Because of the rapid extrusion of the lower and upper middle LST groups (Diehl and Haig 1994), some lava flows could have recorded the same vector of the geomagnetic field therefore artificially increasing the weight of a corresponding direction in statistical analyses. We tested the site-mean directions within each of our four directional groups for serial correlation using the approach of McElhinny et al. (1996) in which the angular separation between two successive directions is compared with the average α_{95} value (α_{95_aver}) for the directional group. First, to account for the dependence of an *i*th site-mean α_{95i} on the number of samples N_i, the angular fraction of the α_{95i} per sample (α_{95i}/N_i) for each site was calculated. The normalized site-mean $\alpha_{95_norm_i}$ was then calculated for each site using

$$\alpha_{95_norm_i} = \left(\frac{1}{B}\sum_{i=1}^{B}\frac{\alpha_{95i}}{N_i}\right)N_i \quad (1)$$

where B is the total number of sites in the directional group. The α_{95_aver} was then computed using

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Table 2. Paleomagnetic data after combining serially correlated directions.

Site	N	D_g (°)	I_g (°)	D_s (°)	I_s (°)	α_{95} (°)	k	P_{lat} (°N)	P_{long} (°E)	A_{95} (°)	K
CS sites (Location 2)											
CS1	9	267.9	46.8	300.3	34.1	3.8	240	34.0	172.5	—	—
CS2	6	261.1	47.2	297.0	37.9	2.7	803	33.6	177.4	—	—
Mean	—	264.5	47.1	298.7	36.0	10.1	609	33.8	174.9	8.8	810
Lower middle LST sites (Location 1)											
L1 (LST1–LST6)	44	283.3	36.8	301.5	36.6	1.2	293	36.0	173.4	—	—
L2 (7)	10	278.7	33.5	295.5	35.6	2.0	582	31.5	177.5	—	—
L3 (8)	8	282.6	34.8	299.7	35.2	3.9	207	34.1	174.0	—	—
L4 (9)	7	283.3	26.6	295.8	27.6	3.9	206	28.1	173.2	—	—
L5 (10)	7	286.0	31.3	300.7	30.7	2.1	863	32.7	170.8	—	—
Mean	—	282.8	32.6	298.6	33.2	4.3	325	32.5	173.8	3.4	503
SI sites (Location 2)											
S1 (SI1)	9	239.2	43.5	280.4	48.1	3.0	282	27.7	195.8	—	—
S2 (SI2)	6	237.2	37.8	272.4	45.8	9.0	36	21.2	199.5	—	—
S3 (SI3)	6	243.9	40.5	279.9	43.5	4.0	281	24.9	193.1	—	—
S4 (SI4, SI5)	14	238.7	40.2	276.1	46.4	2.5	220	24.0	197.5	—	—
S5 (SI6)	7	240.8	32.7	269.8	40.3	2.5	738	16.6	198.1	—	—
S6 (SI7)	4	247.5	43.4	285.3	43.1	6.2	221	28.3	189.1	—	—
S7 (SI8)	6	246.7	37.8	279.0	40.1	3.8	320	22.5	191.7	—	—
S8 (SI9, SI10)	17	239.7	38.6	275.0	44.8	2.2	245	22.4	197.2	—	—
Mean	—	241.7	39.4	277.2	44.1	3.1	259	23.5	195.3	3.4	266
Upper middle LST sites (Location 1)											
U1 (LST12)	6	273.5	34.2	291.2	38.3	4.3	153	29.9	182.3	—	—
U2 (LST13)	8	266.3	35.5	285.3	42.4	4.5	155	27.9	189.1	—	—
U3 (LST14–LST19)	44	262.3	33.9	280.3	42.6	1.6	184	24.7	192.6	—	—
U4 (LST22, LST23)	14	258.2	31.4	274.6	42.0	2.4	247	20.6	196.1	—	—
U5 (LST24)	6	257.0	24.2	269.4	36.0	2.5	481	14.2	196.5	—	—
U6 (LST25)	6	260.6	34.6	279.1	43.9	6.0	104	24.6	194.2	—	—
U7 (LST26)	8	256.1	30.4	271.9	41.9	5.5	104	18.8	197.9	—	—
U8 (LST28, LST29)	17	261.7	28.7	276.6	38.2	2.1	256	20.0	192.6	—	—
U9 (LST30)	6	255.9	24.4	268.4	36.6	5.1	144	13.8	197.5	—	—
U10 (LST31)	8	258.8	29.6	274.2	40.2	3.9	208	19.4	195.4	—	—
Mean	—	260.9	30.8	277.0	40.4	3.7	174	21.4	193.6	4.1	139
Directions from Diehl and Haig (1994)											
LT1	7	295.5	37.9	295.4	12.9	2.4	656	21.9	166.8	—	—
LT2	7	277.1	46.4	281.8	22.3	3.4	322	16.5	180.9	—	—
O1 (OT1)	8	278.8	33.1	281.3	8.9	6.1	84	10.9	176.3	10.9	—
O2 (OT2–OT6)	34	274.0	26.9	276.3	3.3	1.8	174	5.5	178.0	5.5	—
O3 (OT7)	7	281.7	34.7	283.9	10.2	5.8	84	13.2	174.8	13.2	—
O4 (OT8)	8	272.8	29.2	275.6	5.8	2.1	863	5.9	179.4	5.9	—
Mean (OT)	—	276.7	31.0	279.3	7.1	5.7	259	8.9	177.1	4.9	358

Note: N is the number of samples per site; D_g, I_g and D_s, I_s are the site-mean declination and inclination in geographic and stratigraphic coordinates, respectively; α_{95} is the 95% radius of confidence; k is the Fisher (1953) precision parameter; P_{lat} and P_{long} are the latitude and longitude of a virtual geomagnetic pole or a group-mean paleomagnetic pole. A_{95} and K are the 95% radius of confidence and the precision parameter for the group-mean paleomagnetic pole, respectively. Dip azimuth/angle used for structural correction for the structural tilt are 22°/24° for the middle LST flows at the tip of the Keweenaw Peninsula (Location 1) and 356°/38° for the flows of Silver Island (SI) and the mainland (CS) (Location 2).

$$\alpha_{95_aver} = \frac{1}{B}\sum_{i=1}^{B}\alpha_{95_norm_i} \quad (2)$$

The two successive directions were considered serially correlated if the angle between them did not exceed the critical angle $\alpha_{crit} = \alpha_{95_aver} + 0.1°$. The calculated values of α_{crit} are 3.4° for the CS group, 5.0° for the SI group, and 3.3° and 4.4° for the lower and upper middle LST groups at Location 1.

All groups of directions, except for the CS sites, appear to contain serially correlated directions. For further analyses, the correlated directions were averaged. The resulting number of independent field readings is $N = 5$ and $N = 10$ for the lower and upper middle LST at Location 1, and $N = 8$ for the upper middle LST at Location 2 (SI sites; Table 2).

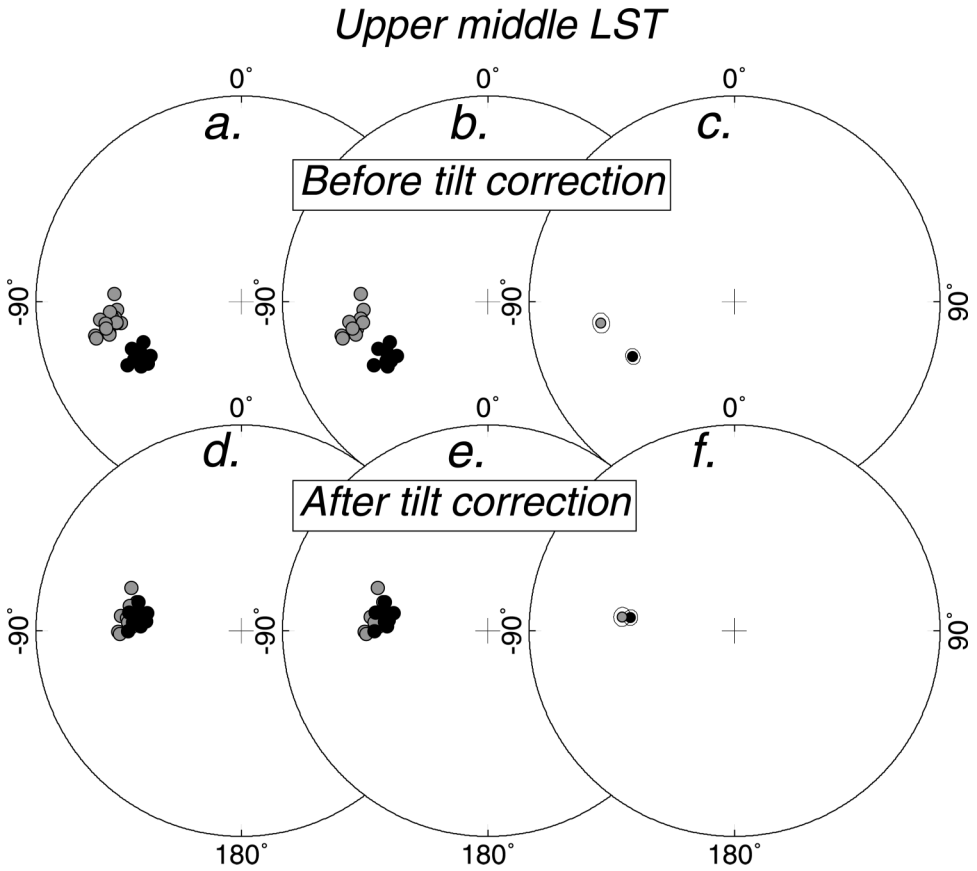
The group mean based on "independent" directions for the lower middle LST ($D = 298.6°$, $I = 33.2°$, $\alpha_{95} = 4.3°$, $N = 5$) remains to be statistically indistinguishable from the CS mean (the angular separation between the group means is 2.8°) (Figs. 5*e*, 5*f*). The new group means for the SI section ($D = 277.2°$, $I = 44.1°$, $\alpha_{95} = 3.1°$, $N = 8$) and the upper middle LST ($D = 277.0°$, $I = 40.4°$, $\alpha_{95} = 3.7°$, $N = 10$) also become statistically indistinguishable (being separated by 3.7°) (Figs. 6*e*, 6*f*).

For consistency, the serial correlation analysis was also applied to eight flows of the outer LST (Diehl and Haig 1994), resulting in four independent directions. The directions of the western middle LST sites (LT1, LT2) from the same study were found to be independent.

The statistical similarity between the group-mean directions based on the independent directions (Table 2) was further tested using the correlation fold test (McFadden 1990). In our discussion

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Fig. 6. Paleomagnetic site-mean directions determined from the upper middle LST sites at Location 1 (grey circles) and the SI sites at Location 2 (black circles) before (*a*, *d*) and after (*b*, *e*) combining serially correlated site-mean directions (see text). (*c*, *f*) The group-mean directions and their respective 95% confidence areas (α_{95}) based on the data in *b* and *e* (Table 2). The data are plotted (equal area projection) before (*a*–*c*) and after (*d*–*f*) correction for structural tilt.



in the following text, we follow the McFadden's (1990) definition of a limb as an area that underwent no internal deformation and can be restored to its initial undeformed attitude by a single tectonic correction. Under this definition, our Locations 1 and 2 represent two different limbs.

In the correlation fold test, a calculated statistic ξ is compared with the tabulated critical value for the total number of sites on both limbs (*N*). If ξ exceeds the critical value, than the null-hypothesis that the paleomagnetic directions do not correlate with tectonic in unfolded position is rejected. McFadden (1990) describes two ways (or test definitions) to calculate ξ, depending on if the angle between the overall in situ and unfolded mean directions (γ_m) is small (definition 1) or large (definition 2) with respect to the angular movement of individual directions ($\{\gamma_i\}$) upon tilt correction (McFadden 1990). Both definitions use the same critical values of ξ. In our data, the angular motion of the mean and individual directions are comparable in magnitude, however the relationship between γ_m and $\{\gamma_i\}$ is opposite for the data from Location 1 ($\gamma_m < \{\gamma_i\}$) and Location 2 ($\gamma_m > \{\gamma_i\}$) within both lower and upper middle LST packages. Therefore, in our analyses, we applied both test definitions, denoting the corresponding statistics ξ_1 and ξ_2. The minimum values of both ξ_1 and ξ_2 were always achieved upon 100% unfolding.

For the lower middle LST, both $\xi_1 = 0.178$ and $\xi_2 = 0.837$ do not exceed the critical value of 3.086 ($N = 7$) at 95% confidence indicating the positive fold test. The maximum *k* statistic (McFadden and Jones 1981) was achieved at 100% unfolding. For the upper middle LST, the test statistics $\xi_1 = 1.943$ and $\xi_2 = 4.056$ are also below the critical value of 4.940 ($N = 18$), and the maximum *k* statistic (McFadden and Jones 1981) is achieved at 100% unfolding, suggesting that ChRM was acquired by these rocks prior to their tectonic deformation.

Discussion

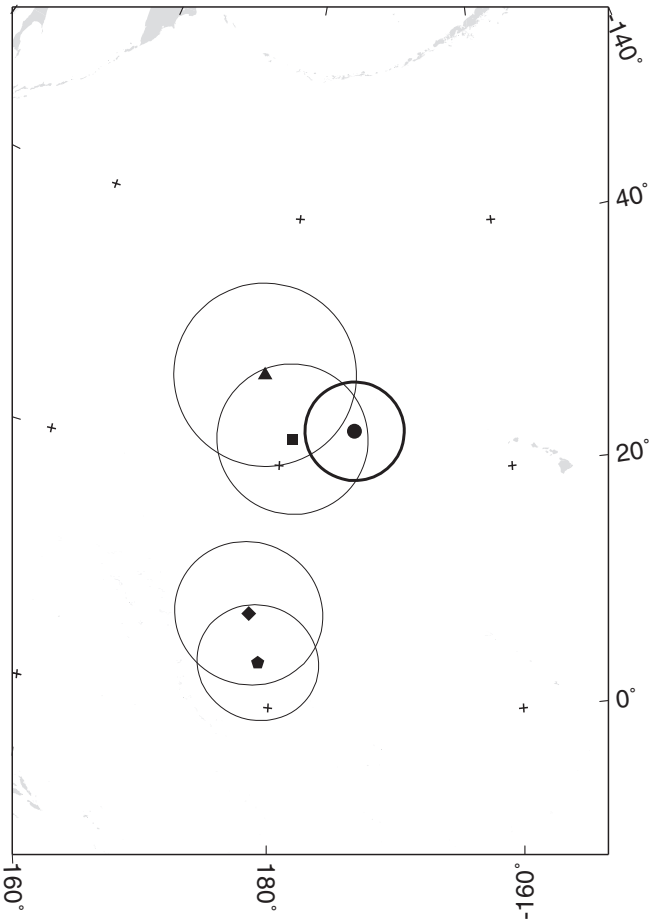
The well-defined ChRM directions obtained from nine additional flows of the upper middle LST at Location 1 are in perfect agreement with the directions reported by Diehl and Haig (1994) (Table 1). The new upper middle LST group mean for Location 1 ($D = 277.8°$, $I = 41.0°$, $\alpha_{95} = 2.3°$, $N = 17$) is statistically indistinguishable (the angular separation is 2.5°) from the previous mean ($D = 279.3°$, $I = 38.7°$, $\alpha_{95} = 4.9°$, $N = 9$) from Diehl and Haig (1994).

Based on the statistical similarity of tilt-corrected paleomagnetic directions within both the lower and upper middle LST stratigraphic packages, we conclude that the 10 lava flows of Silver Island (SI sites) and two flows on the mainland (CS sites) at Location 2 belong to the upper and lower middle LST packages exposed at Location 1, respectively. Furthermore, because the primary origin of ChRM in the lower and upper middle LST at Location 1 has been established by Palmer et al. (1981) and Diehl and Haig (1994), we conclude that the ChRM in all 12 flows at Location 2 is also primary. The primary origin of the ChRM at both locations is further confirmed by our positive correlation fold test. We also note that none of the SI and CS flows yielded a shallow inclination magnetization component observed in the nearby sites of middle and outer LST west of Location 2 (Halls and Palmer 1981; Diehl and Haig 1994).

The new paleomagnetic pole calculated from 31 virtual geomagnetic poles (VGPs) from the middle and outer LST, after combining the serially correlated paleomagnetic directions (Table 2), is lo-

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Fig. 7. Selected paleomagnetic poles from Keweenawan units: the Lake Shore Traps (circle, this study; square, Diehl and Haig 1994); the Portage Lake Volcanics (triangle, calculated from the data by Hnat et al. 2006); the Nonesuch Shale (diamond) and Freda sandstone (pentagon) recalculated from Henry et al. (1977) using site-mean virtual geomagnetic poles.



cated at 23.1°N, 186.4°E (95% radius of confidence for the paleomagnetic pole, A_{95} = 4.0°; N = 31). We note that this location is indistinguishable from that of a mean pole (23.1°N, 185.9°E, A_{95} = 3.3°, N = 49) calculated without accounting for the serial correlation (Table 1). The new pole is similar to the paleopole reported by Diehl and Haig (1994) (P_{lat} = 22.2°N, P_{long} = 181.0°E, A_{95} = 6.1°) (Fig. 7). However, the pole is now based on almost entire sequence of the LST flows and merits six-point classification on Van der Voo's (1990) scale of paleomagnetic reliability. All the reliability criteria, except for the presence of geomagnetic reversals, are satisfied.

The angular dispersion for the virtual geomagnetic poles (Table 2) was calculated using

$$(3) \qquad S^2 = \frac{1}{N-1}\sum_{i=1}^{N}\Delta_i^2$$

where N is the number of individual VGPs and Δ_i is the angle between the ith VGP and the mean pole. The dispersion value was corrected for within-site dispersion (S_w) related to intrinsic variation within a lava flow and experimental uncertainty using

$$(4) \qquad S_b^2 = S^2 - S_w^2/n$$

where S_b is the true-field (between-site) dispersion, n is the average number of samples per site (Doell 1970). Confidence interval on S was estimated using the N – 1 jackknife method (Efron 1982). The calculated value of S = 12.4 ± 1.0° (N = 31) is within the 9.0°–14.6° range of S values reported from other Keweenawan rock sequences (e.g., Halls and Pesonen 1982) and is statistically similar to the mean value of S = 14.0 ± 1.3° (λ_{paleo} = 15°) calculated for the 5–195 Ma interval (Smirnov et al. 2011). Therefore, we conclude that our paleomagnetic dataset faithfully records the paleosecular variation, and the corresponding pole represents a time-averaged geomagnetic field.

The new paleomagnetic pole for Lake Shore Traps is located near the mean paleomagnetic pole (27.2°N, 178.1°E, A_{95} = 7.4°, N = 31) for the Portage Lake Volcanics calculated from Hnat et al's (2006) data (Fig. 7). This proximity further supports the close emplacement age of the two Keweenawan sequences suggested by geochronological and geological data (e.g., Davis and Paces 1990).

The almost 30° difference between the dip azimuth of the lower and upper middle LST flows exposed along the coast line of the Peninsula at Locations 1 and 2 provides an opportunity to examine whether the curvature of the MCR is a primary, preexisting feature or occurred during or after the rocks acquired their primary magnetization. Although none of the two packages represents a time-averaged field, each of them is likely to record the same geomagnetic field direction at both locations. The statistical similarity of paleomagnetic declinations between the CS and lower middle LST group-mean directions (298.7° ± 12.5° and 298.6° ± 5.1°) as well as between the SI and upper middle LST group-mean directions (277.2° ± 4.3° and 277.0° ± 4.9°) (Table 2) indicates that no vertical axis rotation between Locations 1 and 2 have occurred since these rocks acquired their primary magnetization. Therefore, our data strongly suggest that the arcuate geometry of MCR in the Lake Superior region is a primary feature that likely reflects the geometry of a pre-weakened crust exploited by the rifting process as discussed by Hnat et al (2006).

We further note that because each site in Hnat et al. (2006) represented a different flow, the potential declination–strike correlation in their analysis could have been obscured by the changes in geomagnetic field direction due to secular variation. The effect of secular variation is excluded in our approach in which we compare paleomagnetic directions from equivalent lava flows from two locations that have a significant difference in the structural attitude.

Conclusions

The newly sampled lava flow section of the Lake Shore Traps exposed at Silver Island and the adjacent mainland of the Keweenaw Peninsula (Michigan) yielded a pristine record of the Earth's magnetic field directions that existed during their formation. Based on similarity of respective paleomagnetic group-mean directions supported by positive fold tests, we conclude that 10 lava flows exposed at Silver Island and two lava flows exposed on the mainland are equivalent to the lava flows of the upper and lower middle LST sections exposed at the tip of the Peninsula, previously studied by Diehl and Haig (1994).

The new paleomagnetic pole for the Lake Shore Traps is close to that previously reported by Diehl and Haig (1994) hence reinforcing the conclusions of their study. However, because of the availability of positive field (fold) tests, the quality factor for the new pole is now increased to Q = 6 (Van der Voo 1990).

The statistical similarity of tilt-corrected group-mean directions calculated for both lower and upper middle LST at sampling Locations 1 and 2, characterized by considerably different flow dip azimuths, indicates that no vertical axis rotation between the locations had occurred since the formation of LST. The new data lend additional support for the pre-tilting age of magnetization in Keweenawan volcanics (Halls and Pesonen 1982). More impor-

tantly, our data provide an unequivocal evidence for the primary origin of the MCR curvature as was previously suggested by the study of the Portage Lake Volcanics by Hnat et al. (2006).

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